

JEE MAIN REVISION MASTER: SETS AND RELATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Equivalence Relations (Reflexive, Symmetric, Transitive)	18 PYQs	Relations
Moderate	Set Operations & Venn Diagram Principle	5 PYQs	Set Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equivalence Relations (Reflexive, Symmetric, Transitive)	$\text{Reflexive: } \forall a \in A, (a,a) \in R, \quad \text{Symmetric: } (a,b) \in R \implies (b,a) \in R, \quad \text{Transitive: } (a,b), (b,c) \in R \implies (a,c) \in R$
Set Operations & Venn Diagram Principle	$n(A \cup B) = n(A) + n(B) - n(A \cap B), \quad n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equivalence Relations (Reflexive, Symmetric, Transitive) (Very High Yield — 18 PYQs)

Core Principle: Formal definitions of binary relations on sets, properties for equivalence, and total number of reflexive relations $2^n(2^n - 1)$.

■ **Caution / JEE Pitfall:** A relation is vacuously transitive if there are no pairs (a,b) and (b,c) with distinct elements to contradict transitivity.

■ **Speed Trick:** Total number of relations on set of n elements is 2^{n^2} ; total number of symmetric relations is $2^{n(n+1)/2}$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $n(A)=2$ and the number of relations from set A to set B is 1024, then $n(B)$ is..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "If $A=\{x:x \text{ is an integer and } x^2-9=0\}$, $B=\{x:x \text{ is a natural number and } 2 \leq x \leq 5\}$, $C=\{x:x \text{ is a prime number } \leq 4\}$. Then $(B-C) \cup A$..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A and B are two sets having 3 and 6 elements respectively. Consider the following statements. Statement (I): Minimum number of elements ..." [Ans: N/A]

3.2 Set Operations & Venn Diagram Principle (Moderate Yield — 5 PYQs)

Core Principle: Inclusion-exclusion principle for cardinality, De Morgan's set identities, and power set cardinality 2^n .

■ **Caution / JEE Pitfall:** If set A has n elements, the number of proper subsets is $2^n - 1$ (excluding the set itself).

■ **Speed Trick:** Number of elements belonging to EXACTLY two of three sets: $\sum n(A \cap B) - 3n(A \cap B \cap C)$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Let R be the relation in the set ■ given by $R = \{(a,b) : a = b - 2, b > 6\}$. Which of the following is the correct answer?..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "If $A = \{1, 2, 3, 4, \dots, 10\}$, then the number of non-empty subsets of A containing only even number is..." **[Ans: A]**
- **[2026 Shift] Q#4 (MCQ):** "If $A = \{a, b, c, d, e, f\}$, then the number of subsets of A which contains at least 2 elements is..." **[Ans: C]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Let R be the relation in the set ■ given by $R = \{(a,b) : a = b - 2, b > 6\}$. Which of the following is the correct answer?

- (A) $(2, 4) \in R$
- (B) $(3, 8) \in R$
- (C) $(6, 8) \in R$
- (D) $(8, 7) \in R$

Correct Answer: None

Solution Analysis: For a pair (a,b) to be in the relation R, it must satisfy both conditions: $a = b - 2$ and $b > 6$. Check the options: For (2,4) : $2 = 4 - 2$ This is true, but $4 > 6$ is false. So $(2, 4) \notin R$. For (3,8) : Here $b = 8$, so $b > 6$ is true. But $b - 2 = 8 - 2 = 6$ So a should be 6, not 3. Hence $(3, 8) \notin R$. For (6,8) : Here $b = 8$, so $b > 6$ is true. Also, $b - 2 = 8 - 2 = 6$ So $a = 6$. Both conditions are satisfied. Thus, $(6, 8) \in R$ So the correct answer is: Option C

Benchmark Problem 2: KCET 2026 (Q#2)

If $n(A) = 2$ and the number of relations from set A to set B is 1024, then $n(B)$ is

- (A) 2
- (B) 5 **[Correct]**
- (C) 5^2
- (D) 2^5

Correct Answer: B

Solution Analysis: The number of relations from set A to set B is given by $2^{n(A) \times n(B)}$. Here, $n(A) = 2$ and the number of relations is 1024. So, $2^2 \times n(B) = 1024$. Now, $1024 = 2^{10}$. So, $2^{2n(B)} = 2^{10}$. Since the bases are the same, $2n(B) = 10$, $n(B) = 5$. So, the correct option is 5 Option B.